Project Proposal: Music Recommendation and Preference Analysis System

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1. Introduction

This project aims to develop an intelligent Music Recommendation and Preference Analysis System. The system is designed to analyze user preferences based on specific song features such as genre and energy levels to suggest music that aligns with individual tastes, ensuring a personalized music discovery experience.

2. Objectives

- To analyze user listening preferences through mood and genre-based interactions.
- To study song features (Energy, Genre, Artist) to provide accurate recommendations.
- To provide a personalized discovery interface that adapts to the user's selected criteria.
- To implement a secure user environment for tracking individual preference patterns.

3. Key Features

- **Preference Analysis**: The system categorizes music based on attributes like "Energy" (ranging from 0.0 to 1.0) to match the user's current psychological state or mood.
- **Feature-Based Mapping**: Uses metadata analysis (Title, Artist, Genre, and Energy) to filter and suggest the most relevant tracks from the database.
- **Interactive Discovery**: A search-integrated dashboard that allows users to explore music patterns based on specific keywords.
- **Dynamic User Sessions**: Personalized "Welcome" dashboard that tracks the logged-in user's current session and activities.
- **Keyword-Based Search Functionality:** A robust search feature that allows users to find music by analyzing song titles and artist names, providing an alternative to the mood-based discovery.

4. System Logic & Methodology (The "Analysis" Part)

The system follows a Content-Based Filtering approach:

1. Feature Extraction: Songs are stored with specific features (e.g., Pop + High Energy).
2. User Input Analysis: The system captures the user’s choice of mood and genre as "preference data."
3. Pattern Matching: The backend executes a rule-based algorithm to find patterns between the user’s requirements and the song features stored in the SQLite database.
4. Search Pattern Matching: The system utilizes SQL LIKE operators to perform pattern matching on the song database, enabling the retrieval of music that closely matches the user's textual input.

5. Technology Stack

- Backend: Python (Flask Framework) for recommendation logic.
- Frontend: HTML5, CSS3 (Modern Dark UI), and Jinja2.
- Database: SQLite3 for managing song features and user profiles.

6. Conclusion

By analyzing the relationship between musical features and user moods, this system provides a structured approach to music discovery. It moves beyond a simple music player into a preference-based recommendation engine